Waipapa Taumata Rau University of Auckland

COMPSCI 714 - AI Architecture and Design

Equips students with the ability to develop AI applications by introducing well-established AI frameworks and using web-based interactive computing platforms. Students will acquire the skills to implement simple AI techniques using these frameworks and evaluate their performance. Introduces basic practical technologies to investigate artificial intelligence techniques.

Course Details

Course Code	COMPSCI 714
Course Title	AI Architecture and Design
Course Department	Computer Science
Course Level	Stage 7 course
Points	15
Delivery Mode	Campus Experience

Course Offerings

The information below indicates when the course is normally offered. Availability may change up until semester start.

Campus	Semester Usually Offered
City	Semester One

Course Overview

In this course, students will gain knowledge of the steps involved in designing, training and evaluating an AI system. The course contains a large amount of practical work, with regular tutorials and a group project including a hackathon. Students will use well-known frameworks to implement AI applications, mainly based on deep neural networks.

Students will learn how to build AI systems to solve problems using different types of input data. They will follow a pipeline including pre-processing data, choosing an adapted model architecture, making important design choices, tuning hyper-parameters, and evaluating the solution. Students will also be introduced to additional important topics such as transfer learning and generative AI systems.

We want this course to give students relevant practical knowledge and skills to develop their career in AI

research or industry. To that end, student will also attend guest lectures featuring industry and research speakers.

Course Learning Outcomes

CLO #	Outcome	Programme Capability Link
1	Critically evaluate the importance and apply data pre-processing and parsing for machine learning training, as well as identify potential issues of bias and its ethical implications.	
2	Apply and configure an AI environment using well-established AI frameworks and using web-based interactive computing platforms.	
3	Develop, create, and evaluate AI models and identify guidelines for using them.	
4	Apply and evaluate real-world datasets and implement an AI model in Python.	

Additional Information on Assessment

A pass mark in both the practical (assignments and project) and theory (test and exam) components of the course is required to pass the course.

The evaluation of the group project hackathon is done based on a group report and a group presentation.

Assessment Summary

#	Type	%	Classification
1	Coding assignments	30%	Individual Coursework
2	Group project (Hackathon)	30%	Group Coursework
3	Mid-semester test and Final exam	40%	Individual Examination

Assessment Type to Learning Outcome Map

	1	2	3	4
Coding assignments	●	●	●	●
Group project (Hackathon)		●		●
Mid-semester test and Final exam	●		●	

Workload Expectations

This course is a standard 15-point course and students are expected to spend 10 hours per week involved in each 15-point course that they are enrolled in.

In total, expect 150 hours of work over the 12 weeks, broken down into as follows: 36 hours of lectures/tutorials, 48 hours of reading and thinking about the content and 66 hours of work on assignments and/or test preparation.

You can expect a typical week of work to be broken down as follows:

- 3 hours of lecture/tutorial (tutorials will be run at lecture times)
- 4 hours of reading and thinking about the content
- 5.5 hours of work on assignments/projects and/or test preparation

Delivery Mode Details

Attendance is required at scheduled activities including lectures and tutorials to complete components of the course.
Lectures and tutorials will be available as recordings, but attendance is crucial for you to complete the course.
Attendance on campus is required for the test.
The activities for the course are scheduled as a standard weekly timetable.

Key Topics

Some of the key topics taught in this course are:

- Data pre-processing
- AI pipeline

- Basics of neural networks and Deep Neural Networks (DNN) implementation
- DNN architectures to learn with images and sequences
- DNN design choices
- Hyperparameter tuning
- Transfer learning
- Generative AI architectures
- Multimodal AI
- Cloud computing
- Additional topics relevant to AI architecture and design

Special Requirements

Students are expected to attend all in-person activities, including lectures and tutorials. There will be a hackathon event organised in week 10 or 11, as part of the group project. The event will take place on Campus and is likely to be held over a week-end.

Learning Resources

Taught courses use a learning and collaboration tool called Canvas to provide students with learning materials including reading lists and lecture recordings (where available). Please remember that the recording of any class on a personal device requires the permission of the instructor.

Additional Information on Learning Resources

You are strongly recommended to bring a **personal laptop** in tutorials. If you do not own one, laptop loan options are available at the University of Auckland Student IT Hub:https://www.auckland.ac.nz/en/students/student-support/student-it-hub/student-long-term-laptop-loan.html
We do not have a mandatory textbook for the course, but we use a **recommended textbook**:

- Géron, A. (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition. O'Reilly.

Python coding requirements In this course, we will be using Python as programming language to implement and experiment with AI systems. **We assume that students already have some experience with programming**, especially using Python. You should be familiar with Python's main scientific libraries, in particular Numpy., Pandas., and Matplotlib. If you're new to Python or need a refresher before diving into this course, we highly recommend completing the LearnPython tutorial (at least the "Learn the Basics" and "Data Science Tutorials"). Other options are the official tutorial at Python.org or the Learn Python Kaggle tutorial. The advantage of the Kaggle tutorial is that it is using notebooks, which we will use to edit and run code in the tutorials.

Disclaimer

Elements of this outline may be subject to change. The latest information about taught courses is made available to enrolled students in Canvas.

Students may be asked to submit assessments digitally. The University reserves the right to conduct scheduled tests and examinations online or through the use of computers or other electronic devices. Where tests or examinations are conducted online remote invigilation arrangements may be used. In exceptional circumstances changes to elements of this course may be necessary at short notice. Students enrolled in this course will be informed of any such changes and the reasons for them, as soon as possible, through Canvas.